

JEE MAIN REVISION MASTER: STRUCTURE OF ATOM SYLLABUS REDUCED

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 219 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Very High	Structure Of Atom - Core Application	219 PYQs	General Concept

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
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3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 Structure Of Atom - Core Application (Very High Yield — 219 PYQs)

Core Principle: Key problem pattern in Structure Of Atom.

■ ■ Caution / JEE Pitfall: Watch for algebraic and sign errors.

■ Speed Trick: Verify dimensions and boundary conditions.

Representative Source PYQs:

- [2026 Evening Shift] Q#1 (MCQ): "The Bohr radius of a hydrogen like species is 70.53 pm . The species and the stationary state (n) are respectively (Given : Hydrogen atom Bohr radius $a_0 = 0.529 \text{ \AA}$) [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Arrange the following atomic orbitals of multi electron atoms in order of increasing energy. A. $n=3, l=2, m=+1$ B. $n=4, l=0, m=0$ C. $n=6, l=1, m=0$ [Ans: N/A]
- [2026 Morning Shift] Q#1 (MCQ): "Which of the following is correct set of 4 quantum numbers of 19th electron in Chromium (Atomic number =24) in accordance with Aufbau principle [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: JEE Main 2026 (Online) 21st January Evening Shift (Q#1)

Consider the following spectral lines for atomic hydrogen : A. First line of Paschen series B. Second line of Balmer series C. Third line of Paschen series D. Fourth line of Brackett series The correct arrangement of the above lines in ascending order of energy is :

- (A) < C < A < B
- (B) < B < C < D
- (C) < A < C < B **[Correct]**
- (D) < D < B < A

Correct Answer: C

Solution Analysis: For hydrogen spectral lines (emission), $E = \frac{hc}{\lambda} \Rightarrow E \propto \frac{1}{\lambda}$ Using Rydberg formula (NCERT): $\frac{1}{\lambda} = R \left(\frac{1}{n_1^2} - \frac{1}{n_2^2} \right)$ (n₂ > n₁) So photon energy is proportional to: $\Delta E = \frac{1}{n_1^2} - \frac{1}{n_2^2}$ Now identify each line: Paschen series : n₁=3 A (first line) : n₂=4 → 3 C (third line) : n₂=6 → 3 Balmer series : n₁=2 B (second line) : n₂=4 → 2 Brackett series : n₁=4 D (fourth line) : n₂=8 → 4 Compute Δ for each: D: $\frac{1}{8^2} - \frac{1}{4^2}$